Lecture 24

Let's call some SNPs

Th typical variant caller workflow

1. Produce alignments
2. Call variants from the alignments

What if the alignments are "wrong" in some regions?

Wrong means not **mathematically incorrect** - it means that reality does not conform to the scoring matrix that you chose.

You then may need to **inspect**, **correct** or **recalibrate** the alignment or results.

Why did we spend so much time with filtering?

Just about every analysis will depend on the alignment result.

In simple cases using the original BAM file may suffice.

Often it is essential that your input BAM file contains only the data that is appropriate for the analysis.

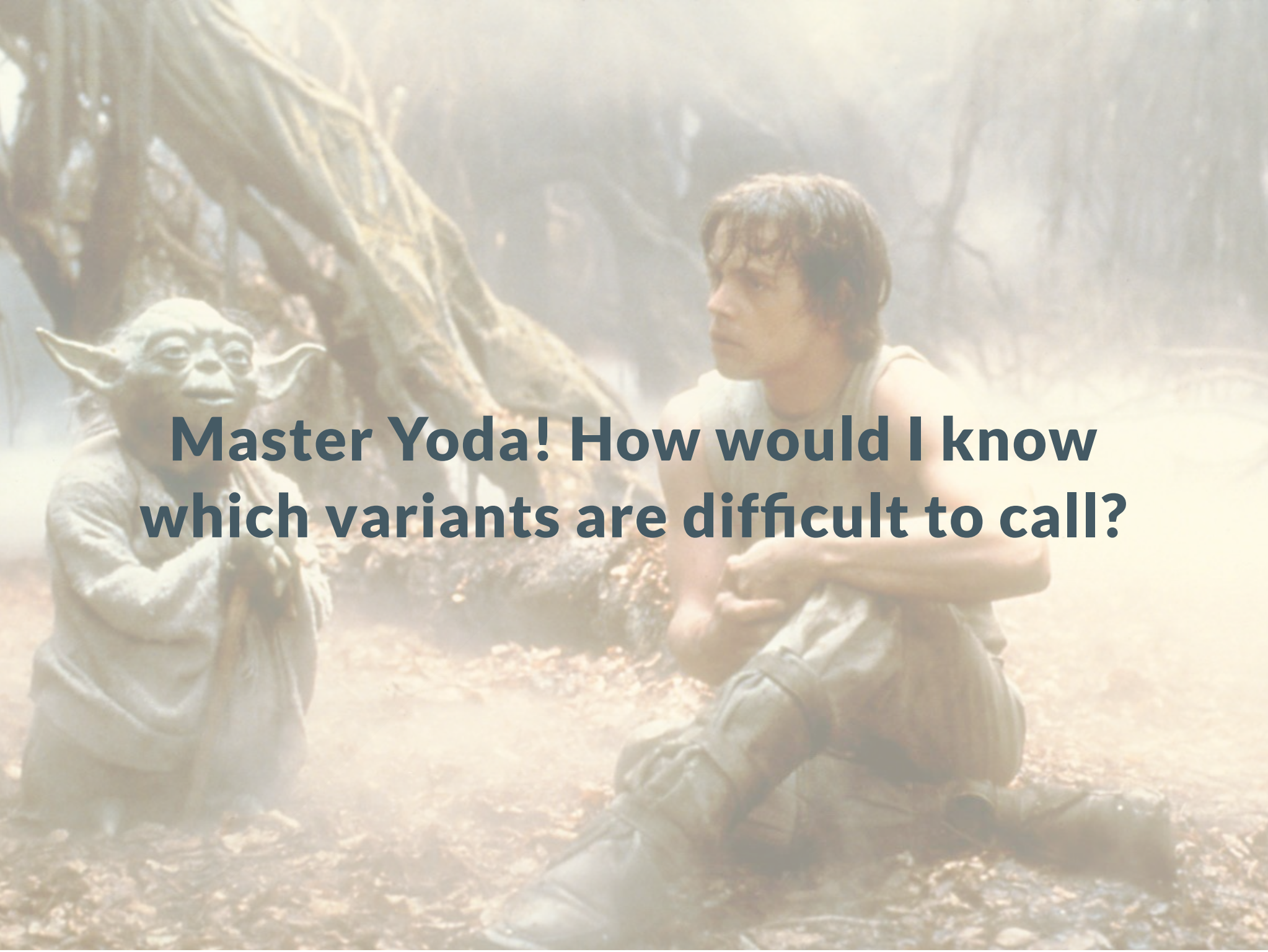
You must not be "afraid" of the BAM file. You are not afraid of it when you know what is in it.

What is the right way to call SNPs?

The answer depends on the complexity of the information in the genome.

There are several tacit and built-in assumptions that each software developer builds into their tool.

- Easy regions: can be solved by any approach
- Not so easy regions: some results are incorrect
- Difficult regions: need manual supervision


Master Yoda! How would I know
which variants are difficult to call?


Hard to call, difficult variants are!

Expectations

For example for human genome

Doing an 80% job (80% of variants are correct and do in fact exist) is "easy." "Easy" means there are documented parameter settings that do a good job.

Doing a 100% job is impossible.

Your results will typically fall somewhere between the two: 80%-100%

Let's call SNPs: Get the data

Build another `megaton.sh` that can run different SNP callers.

Reference genome: `AF086833`

```
REF=db/AF086833.fa

# Make a directory to store the reference.
mkdir -p db

# Get the reference.
efetch -db nuccore -id AF086833 -format fasta > $REF

# Index the reference genome.
bwa index $REF
```

Let's call SNPs: Generate a mutated genome

Simulate reads with known mutations:

wgsim -N 2500 $REF read1.fq read2.fq > mutations.txt

The mutations file has:

```
cat mutations.txt
AF086833.2 46 - G -
AF086833.2 174 GG - -
...
AF086833.2 2791 - GG -
...
```

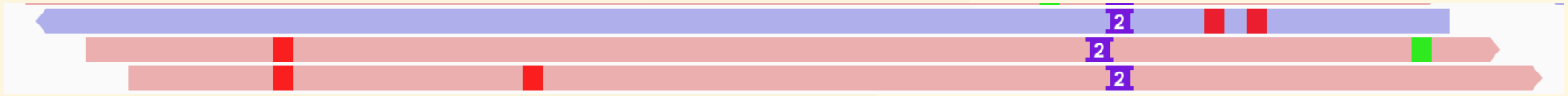
Let's call SNPs: Generate the alignments

Align the reads:

```
bwa mem $REF read1.fq read2.fq | samtools sort > bwa.bam
samtools index bwa.bam
```

What is a misalignment?

Remember that your simulation might be different. Zoom in closer over the two insertions at 2791



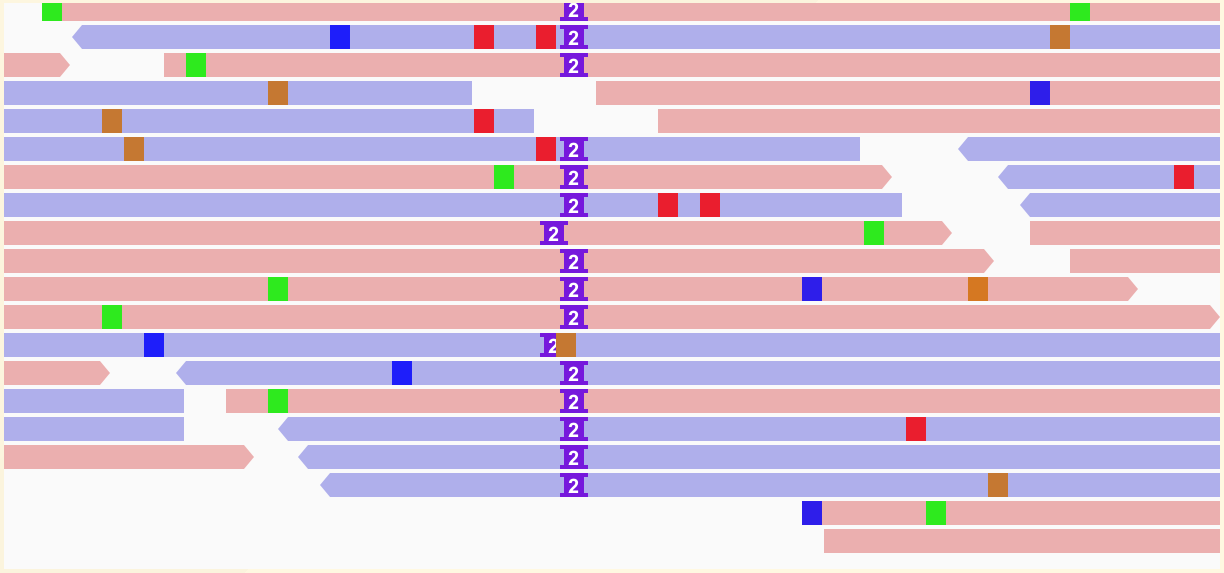
See how the middle read indicates the insertion in the "wrong" position?

Also, note how otherwise it is **perfectly matching** around the INDEL!

It is "misaligned" but not through the fault of the aligner. We could **fix** the middle read by looking at the other reads around it: **realignment**

The Ebola genome is information dense

In this case, there are plenty of "correct" alignments that will "rescue" the incorrect one.



This is not always the case. You could have the majority or reads misalign --> **incorrect calls**

What is complex and what is simple?

In information technology:

- Complex usually means lots of information.
- Simple usually means little information.

In bioinformatics, it is easier to solve genomes with lots of information than genomes with little information.

Low information content means redundant elements, repetitive regions, low complexity base content.

Bioinformatician's inverted the meaning of the word "complex."

What we call "complex" means one that is difficult to solve because the information is spread out across lots of bases.

So the Ebola genome is "simple."

That means it is dense in information, and that makes it less sensitive to scoring matrices.

The more "complex" a genome, the more likely is that the alignments go awry.

Think about it as not having enough bases to **anchor** a read to the right location.

Accurately locating the origin becomes even more challenging when the read does not even match the reference because of the mutation.

Let's call SNPs: how to I pick the right tool?

Different people swear by various tools.

Some tools are tuned to some specific use cases.

`GATK` for human/cancer genome - but it is not a tool - more of a lifestyle - one that happens to be unecessarily complicated and obtuse. You must bend the knee when you use `GATK`.

We will demonstrate a few other approaches:

- `bcftools`
- `freebayes`

Let's call SNPs: using `bcftools`

A two step process:

```
# Copmute the genotype likelihoods for each base.
samtools mpileup -uvf $REF bwa.bam > genotypes.vcf
```

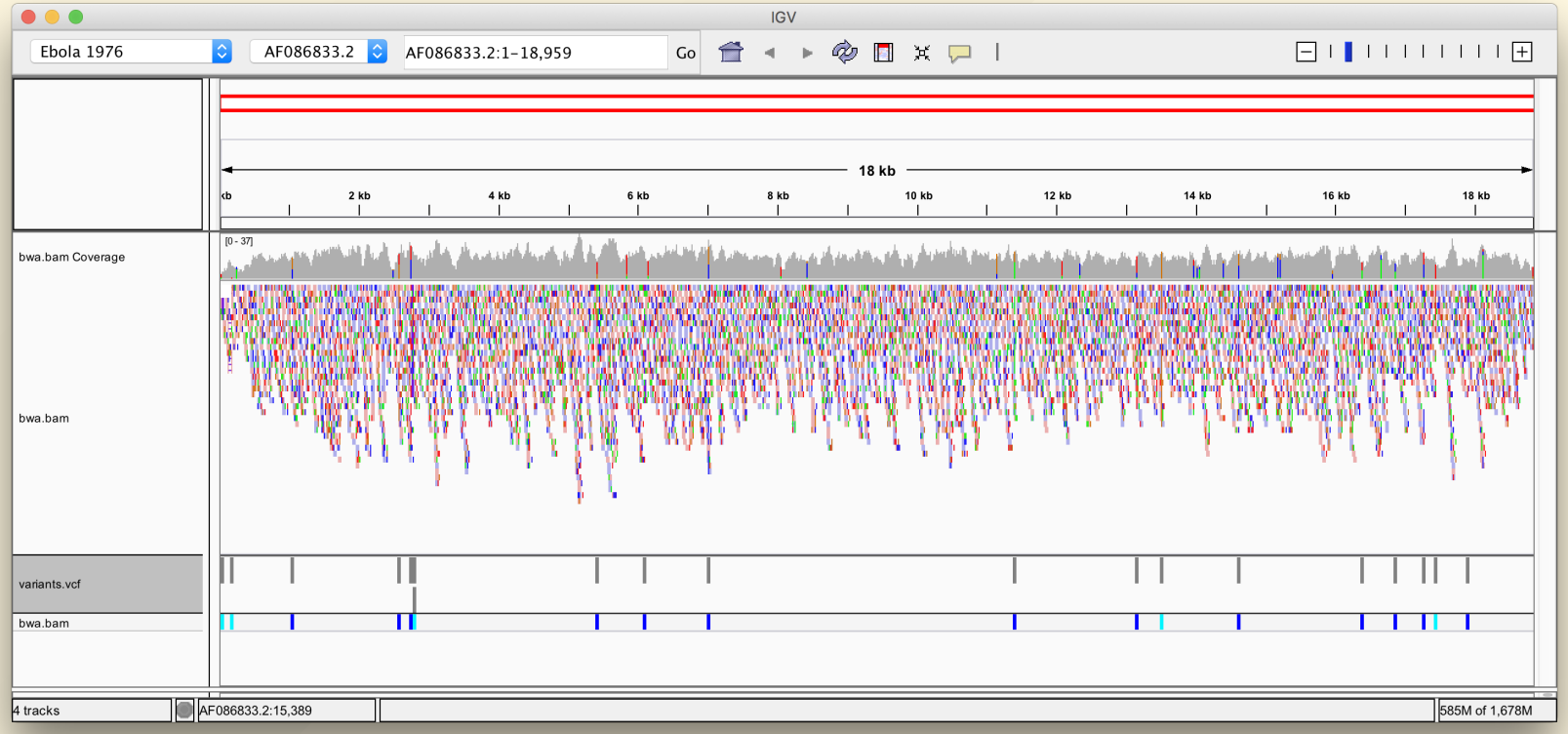
VCF files are explained in other lectures. Just take them as they are here.

```
# Call the variants with bcftools.
bcftools call  -vm -Ov genotypes.vcf > variants.vcf
```

The final results of this sequencing experiment are in `variants.vcf`. This is the file that needs a biological interpretation.

Let's call SNPs: interpreting the output

The bottom tracks are the variant calls:



Let's call SNPs: did it get the funky part right?

Well, ahem...no



Let's call SNPs: using `freebayes`

A different tool, with a better understanding of misalignments:

```
freebayes -f $REF bwa.bam > variants2.vcf
```

Let's call SNPs: visualizing `freebayes`

Freebayes has called the tricky variation correctly



Observations

This does not necessarily mean that `freebayes` is always better.

Tools built for a specific task usually perform better - their developers understand the pitfalls of that job and have builtin corrections.

All tools will fail in some cases.

Warning sign: variants close by one another (10-30 bases or fewer)